Tutorial -5

1. **Write a C# Sharp program that stores elements in an array and prints them.**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_5
{
    internal class pROGRAM1
    {
        public static void Main(string[] args)
        {
            int[] arr = new int[5];
            for (int i = 0; i < arr.Length; i++)
            {
                Console.WriteLine("Enter number " + (i + 1) + ":");
                arr[i] = Convert.ToInt32(Console.ReadLine());
            }


            for (int i = 0; i < arr.Length; i++)
            {
                Console.WriteLine(arr[i]);
            }
            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-5>pROGRAM1
Enter number 1:
3
Enter number 2:
2
Enter number 3:
4
Enter number 4:
5
Enter number 5:
5
3
2
4
5
5
```

2. Write a program of sorting an array. Declare single dimensional array and accept 5 integer values from the user. Then sort the input in ascending order and display output.

```
3. using System;
4. using System.Collections.Generic;
5. using System.Linq;
6. using System.Text;
7. using System.Threading.Tasks;
8.
9. namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_5
10.{
11.    internal class Short_array
12.    {
13.        public static void Main(string[] args)
14.        {
15.            short[] arr = new short[5];
16.            for (int i = 0; i < arr.Length; i++)
17.            {
18.                Console.WriteLine("Enter number " + (i + 1) + ":");
19.                arr[i] = Convert.ToInt16(Console.ReadLine());
20.            }
21.            Array.Sort(arr);
22.            for (int i = 0; i < arr.Length; i++)
23.            {
24.                Console.WriteLine(arr[i]);
25.            }
26.            Console.ReadLine();
27.
28.
29.
```

```
30.         }
31.     }
32.}
33.
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-5>Short_array
Enter number 1:
3
Enter number 2:
4
Enter number 3:
5
Enter number 4:
2
Enter number 5:
3
2
3
3
4
5
```

3. Write a C# Sharp program to read n values in an array and display them in reverse order.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_5
{
    internal class reverse
    {
        public static void Main(string[] args)
        {
            int[] arr = new int[5];
            for (int i = 0; i < arr.Length; i++)
            {
                Console.WriteLine("Enter number " + (i + 1) + ":");
```

```
            arr[i] = Convert.ToInt32(Console.ReadLine());
        }
        Array.Reverse(arr);
        for (int i = 0; i < arr.Length; i++)
        {
            Console.WriteLine(arr[i]);
        }
        Console.ReadLine();
    }
  }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-5>reverse
Enter number 1:
2
Enter number 2:
3
Enter number 3:
5
Enter number 4:
2
Enter number 5:
4
4
2
5
3
2
```

4. Write a C# Sharp program to copy the elements of one array into another array.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_5
```

```
{
    internal class copy1
    {
        public static void Main(string[] args)
        {
            int[] arr1 = new int[5];
            for (int i = 0; i < arr1.Length; i++)
            {
                Console.WriteLine("Enter number " + (i + 1) + ":");
                arr1[i] = Convert.ToInt32(Console.ReadLine());
            }
            int[] arr2 = new int[arr1.Length];
            Array.Copy(arr1, arr2, arr1.Length);
            for (int i = 0; i < arr2.Length; i++)
            {
                Console.WriteLine(arr2[i]);
            }
            Console.ReadLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-5> copy1
Enter number 1:
3
Enter number 2:
5
Enter number 3:
2
Enter number 4:
1
Enter number 5:
1
3
5
2
1
1
```

5. Write a C# Sharp program to count duplicate elements in an array.

```
using System;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_5
{

    interface Count_duplicate
    {



        public static void Main()
        {
            int[] arr = { 10, 20, 10, 30, 20, 40, 10 };
            for (int i = 0; i < arr.Length; i++)
            {
                int count = 0;
                for (int j = 0; j < arr.Length; j++)
```

```
            {
                if (arr[i] == arr[j])
                    count++;
            }
            if (count > 1)
                Console.WriteLine(arr[i] + " occurs " + count + "times");
        }
        Console.Read();
    }
  }
}
```

6. Write a C# Sharp program to find the maximum and minimum elements in an array.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_5
{
    internal class min_max
    {



        static void Main()
        {
            int[] arr = { 10, 25, 5, 40, 15 };

            int max = arr[0];
```

```
        int min = arr[0];

        for (int i = 1; i < arr.Length; i++)
        {
            if (arr[i] > max)
                max = arr[i];

            if (arr[i] < min)
                min = arr[i];
        }

        Console.WriteLine("Maximum element = " + max);
        Console.WriteLine("Minimum element = " + min);
    }
  }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-5>min_max
Maximum element = 40
Minimum element = 5
```

7. Write a program in C# Sharp to separate odd and even integers into separate arrays.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_5
{
    internal class Odd_even_sepatrete
```

```
    {
        static void Main()
        {
            int[] arr = { 10, 25, 5, 40, 15 };
            int[] odd = new int[arr.Length];
            int[] even = new int[arr.Length];
            int o = 0, e = 0;
            Console.WriteLine("Odd numbers in the array:");
            foreach (int num in arr)
            {
                if (num % 2 != 0)
                {
                    even[o] = num;
                    o++;
                    Console.WriteLine(num);
                }
            }
            Console.WriteLine("Even numbers in the array:");
            foreach (int num in arr)
            {
                if (num % 2 == 0)
                {
                    even[e] = num;
                    e++;
                    Console.WriteLine(num);
                }
            }
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-5>Odd_even_sepatrete
Odd numbers in the array:
25
5
15
Even numbers in the array:
10
40
```

8. Write a C# Sharp program to sort array elements in descending order.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_5
{
    internal class Desending_order
    {
        static void Main()
        {
            int[] arr = { 10, 25, 5, 40, 15 };
            Array.Sort(arr);
            Array.Reverse(arr);
            Console.WriteLine("Array in descending order:");
            foreach (int num in arr)
            {
                Console.Write(num + " ");
            }
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-5>Desending_order
Array in descending order:
40 25 15 10 5
```

9. Write a C# Sharp program to delete an element at the desired position from an array.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_5
{
    internal class del_position
    {
        static void Main()
        {
            int[] arr = { 10, 25, 5, 40, 15 };
            int position = 2; // Position to delete (0-based index)
            if (position < 0 || position >= arr.Length)
            {
                Console.WriteLine("Invalid position.");
                return;
            }
            for (int i = position; i < arr.Length - 1; i++)
            {
                arr[i] = arr[i + 1];
            }
            Array.Resize(ref arr, arr.Length - 1); // Resize the array to remove the
last element
            Console.WriteLine("Array after deleting element at position " +
position + ":");
```

```
        foreach (int num in arr)
        {
            Console.Write(num + " ");
        }
    }

}
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-5>del_position
Array after deleting element at position 2:
10 25 40 15
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-5>
```

10. Write a C# Sharp program for adding two matrices of the same size.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace _24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL_5
{
    internal class add_matrix
    {
        static void Main()
        {
            int[,] matrix1 = { { 1, 2, 3 }, { 4, 5, 6 }, { 7, 8, 9 } };
            int[,] matrix2 = { { 9, 8, 7 }, { 6, 5, 4 }, { 3, 2, 1 } };
            int rows = matrix1.GetLength(0);
            int cols = matrix1.GetLength(1);
            int[,] result = new int[rows, cols];
            for (int i = 0; i < rows; i++)
            {
                for (int j = 0; j < cols; j++)
```

```
            {
                result[i, j] = matrix1[i, j] + matrix2[i, j];
            }
        }
        Console.WriteLine("Resultant Matrix after Addition:");
        for (int i = 0; i < rows; i++)
        {
            for (int j = 0; j < cols; j++)
            {
                Console.Write(result[i, j] + " ");
            }
            Console.WriteLine();
        }
    }
}
```

```
D:\.net Project\24SOECE11019_RAJVI_LUNAGARIYA_TUTORIAL-5>add_matrix
Resultant Matrix after Addition:
10 10 10
10 10 10
10 10 10
```